Testing for Nutrients

There are some simple tests to determine the nutrients in food. You are going to carry out some of these tests, then test various food samples for nutrients

Part A: Testing for sugars

Equipment: 2 test tubes, labels, glucose solution, distilled water, Benedict's solution, 250 mL beaker, Bunsen burner, tripod and gauze mat.

Procedure:

1. Label two test tubes (A and B) and place them in a test tube rack.
2. Add 5 mL of glucose solution to test tube A.
3. Add 5 mL of distilled water to test tube B.
4. To each test tube, add 5 drops of Benedict's Solution
5. Place the two test tubes in a 250 mL beaker containing 150 mL of water.
6. Place the beaker on a wire mat over a tripod and heat with a Bunsen burner.
7. Bring the water to the boil and leave the test tubes in the water until a change occurs.
8. Record your results below.

Results:

	Contents of tube	Observations
Test tube A	Glucose	
Test tube B	Water	

Conclusion:

Part B: Testing for fats

Equipment: Brown paper, oil, water.

Procedure:

1. Label two pieces of brown paper A and B.
2. Add two drops of oil to the centre of A.
3. Add two drops of water the centre of B.
4. Allow the paper to dry, then hold it up to the light.
5. Lipids leave a transparent mark.
6. Record you results below.

Results:

	Substance on Paper	Observations
Paper A	Oil	
Paper B	Water	

Conclusion:

Part C: Testing for Protein

Equipment: 2 test tubes, labels, protein solution, distilled water, Sodium hydroxide solution, Copper sulphate solution

Procedure:

1. Label the test tubes A and B.
2. Add 10 mL of protein solution to test tube A.
3. Add 10 mL of distilled water to test tube B.
4. To each tube, add 10 drops of sodium hydroxide solution.
5. Tilt the test tube and add two or three drops of copper sulphate solution down the side of the tube.
6. A violet ring at the surface indicates protein.
7. Record your results below.

Results:

	Contents of Tube	Observations
Test Tube A	Protein	
Test Tube B	Water	

Conclusion:

Part D: Testing for starch

Equipment: Two test tubes, labels, starch solution, distilled water, iodine solution.

Procedure:

1. Label the test tubes A and B.
2. In tube A, put 10 mL of starch solution.
3. In tube B, put 10 mL of distilled water.
4. Add 3 drops of iodine solution to each tube.
5. Iodine turns black in the presence of starch.
6. Record your results below.

Results:

	Contents of Tube	Observations
Test Tube A	Starch	
Test Tube B	Water	

Conclusions:

Part E: Testing Foods

Use the steps outlined in parts A to D to identify the presence of nutrients in different food samples.

Record your results in the table below.

Food Tested	Sugar	Fats	Protein	Starch

Questions:

1. Which food had the most different types of nutrients present? Which nutrients did it have?

2. Which food had the least different types of nutrients present? Which nutrients did it have?

3. Which nutrient was in the most foods? Which foods was it in?